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Simulation and Optimization of Microgrid Distributed Generation: a Case Study of University Abdelmalek Essaâdi in Morocco

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Abstract

Renewable energy based Distributed Generation (DG) in form of PV power system can tackle the growing demand of depletion of electrical energy resources and the increasing prices of oil. Smart grid is the innovation of recent infrastructure grid and DG is an essential part of it. The Moroccan government has also made significant changes to its renewable energy policy to promote clean energy. The aim of this paper is to show the way to implement the concept of simulating and optimizing the real time of dynamic data in university Abdelmalek Essaâdi with HOMER as platform software. According to the simulation result, it is seen that microgrid implementation in Morocco is beneficial with regards to both optimum solution and sensitivity analysis.

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Keywords: renewable energy; distributed generation; HOMER; microgrid.

1. Introduction

In developing countries, the demand of energy is expected to grow fastly. The Moroccan government applies a new energy policy to tackle this growing demand like renewable energy resources which provides clean sustainable environment [1, 2].

The future electric grid must be smart, reliable, efficient and enduring to any damages. In general, microgrid can be called as a self –contained between power systems of a number of interconnected loads and resources, the

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characteristics of microgrid contain an intelligent management system that interfaces with equipment by wireless technologies [3-5].

This study is in progress in order to develop a standard procedure for the design of solar photovoltaic (PV) connected to a network and a large scale using the rooftop of the university Abdelmalek Essaâdi. The standard procedure developed will be validated during the design of the system connected to the network. The performance of the proposed microgrid system will also be simulated and optimized using Hybrid Optimization Model for Electric Renewable (HOMER) software.

2. Geographical and parameter coordination of the university Abdelmalek Essaâdi, Morocco

For the implementation of microgrid concept, the chosen site is tested; explored and different specifications are given below in Fig.1. (a) Abdelmalek Essaâdi university is located in the north of Morocco and is situated in a Mediterranean climate at latitude of $35^{\circ}56'$ (north) and longitude of $-5^{\circ}36'$ (east), Fig. 1 (b) depicting both monthly average global solar radiation and clearness index for 2015, the solar radiation peaked between June, July and August. The average is close to 5.11 KWh/day.

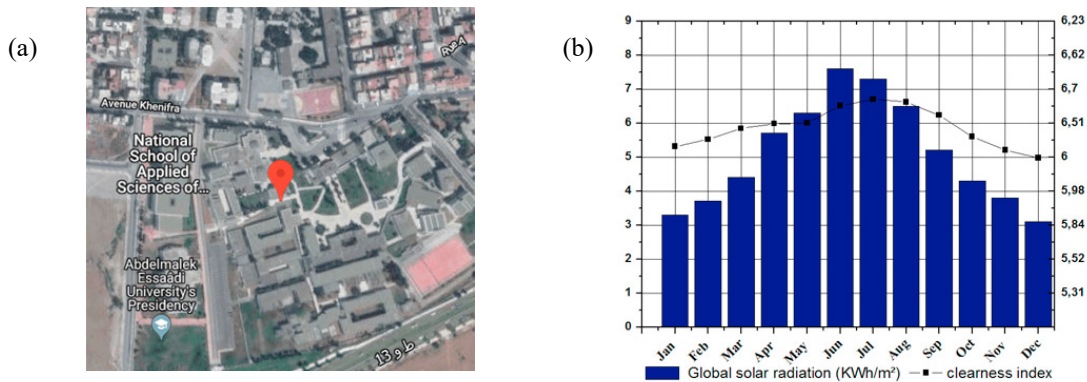


Fig. 1. (a) Location of area; (b) Monthly profile of global solar radiation and clearness index of Tetuan city.

The clearness index is approximately low during October, November and December due to the monsoon season [5]. Fig. 2 presents a typical daily profile for January. As we can see, the load demand becomes high between 6 A.M to 9 P.M during working hours and remains at the minimum level in the early morning and outside working hours [6]. The average energy demand is 10626.76 KWh daily. The peak demand is 35552.38 KW. The varying load profile this year has been presented as 465 KW/day. Therefore, the real time profile of the proposed site is obtained without including the load shedding durations and is considered as input for optimal configuration of microgrid system and Total assumption of designated area.

3. System description

3.1. Microgrid system design

According to the microgrid system design, the optimization size of microgrid based on PV panels system is very important and leads to a good ratio between performances and cost of installations [7,8]. Before that, the load profile and accessible insolation must be calculated. Thus, they are presented in the following sections.

The design of microgrid power system in a specific location depends on many factors such as the capacity storage, electric load (daily, seasonally), renewable resources and technical components (details, cost, controls, type of dispatch strategy, etc.).

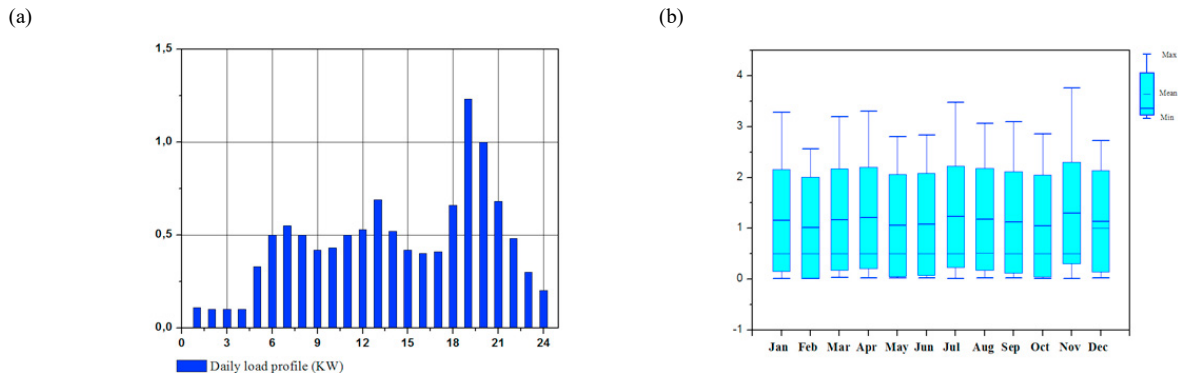


Fig. 2. (a) Hourly load profile; (b) yearly load profile.

The platform HOMER software is a hybrid system for designing, optimizing and analysing developed by National Renewable Energy Laboratory (NREL) to serve the desired loads [9,10].

4. Microgrid optimization platform: HOMER

4.1. Simulation, optimization and sensitivity analysis

HOMER simulation simulates as the first phase the process of the system by making balance computation for thousands of hourly simulation in a year. The optimum design of microgrid system must be performed to ensure the best dispatching between demand and supply. HOMER estimates the lifecycle cost in order to rank this systems. The optimization as the second phase of HOMER process is used to identify and evaluate different scenarios of the system configuration between the lowest and highest Net Present Cost (NPC). As a result, the optimization is classified according to a specific scenario of the inputs of the sensitivity variables.

Table 1. Buy and sell of power grid price (\$/KWh).

Specifications	Cost (\$)
Grid power cost (\$/KWh)	0.16
Grid sellback cost(\$/KWh)	0.08

The last software phase is sensitivity analysis; it is the number of iteration of optimization process for the microgrid system. Therefore, after the process done with all sensitivity variables, the best configuration is tracked from several results, that can tolerate the scenario which characterized by the above mentioned case. As a result, the values are presented from the lowest to the highest NPC which make choice to pick the best one for the optimal system of the study site [11-17].

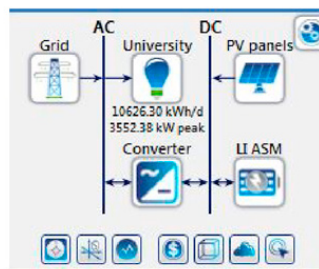


Fig. 3. HOMER microgrid designer.

The planned configuration contains: PV panels, Dc to Ac converter, storage and advance grid. The schematic of this microgrid system is shown in Fig.3. The project lifetimes endure 25 years and discount rate takes 8%. The system performance includes different situations using real weather data (solar radiation) as an input. The implementation study is to verify and determine the most suitable value of each scenario that interests modeller. The scenario includes: The size of the PV array, the Capacity of batteries and the size of the converter.

4.2. Power management dispatching

The microgrid based on PV panel system will be created by HOMER; the rated capacity used includes the efficiency and area of the PV module to feed the load demand for the university. The extra energy is stored in the battery to reach its maximum capacity. The second role of the battery is to import and export the energy when the system faces deficit of power. The converter is used in this system to facilitate the energy transfer between Dc/Ac components. HOMER simulates the advance grid as an important component of the microgrid which can purchase or sell Ac electricity. The cost of purchasing electricity covers both energy costs for a billing period and claimed peak demand charges in the same billing period [18-20]. The electricity sold to the distribution grid in accordance with the price indicated in Table.1 .The details of the planned microgrid system components describe in Table.2.

Table 2. Microgrid system details.

System data	
Project life time	25 years
Operating strategy	HOMER Load following
discount rate	8%
PV array	
Technology	TSM-PD14 320-335w
Peak power (KWp)	0.320
Vmpp (V)	37.1
Immp	8.63
Efficiency (%)	74%
Capital cost (\$US/KWpk)	1700.2
O & M cost (\$US/KWpk)	850.1
Life time	25 years
Number unit of basic PV unit	0, 12,25,45,50
Battery	
Technology	Li-ASM
Capacity (KWh)	1
Nominal capacity (KWh)	1.02
Voltage (V)	12
Capital cost (\$US)	3500
O & M cost (\$US)	1570
Life time	10 years
Number unit of batteries which can be considered	10,200,,2000,8000,16000
Converter	
Technology	Leonics GTP507
Capacity (KWh)	125
Capital cost (\$US)	300
replacement cost (\$US)	300

Life time	15
Efficiency (%)	96
Number unit of converter which can be considered	0,8,16,24,32,125

5. Simulation results

5.1. Optimal configuration and cost components

As already explained, the optimal microgrid design for various combination cases are obtained from HOMER simulation using the parameters described in section 3. The Comparison of the electric energy production and the consumption for several microgrid configurations where the corresponding details of the optimal microgrid plan for each case is conducted and presented in Table. 3-4 and Fig.5-6. The NPC of the first case without storage is 9.39 M\$. The energy production is 3929.86 KWh/ year. The second case completely relies on solar PV panels, battery storage, converter and grid. The energy production is close to 3929.86 KWh/ year and the NPC is 9.41 M\$. The energy productions of the third and fourth cases without solar PV panels are 3878.6 KWh/year.

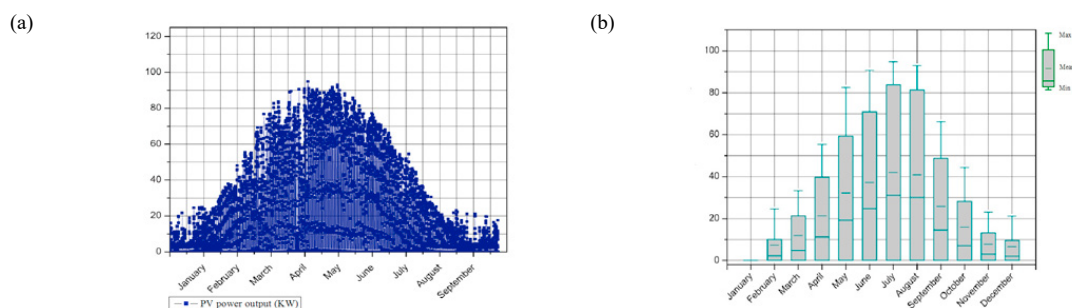


Fig. 4. (a) PV panels power output evolutions over a 24hours period (kW); (b) Monthly average renewable energy output.

This work aims to find the lowest cost microgrid configuration. HOMER software eliminates insignificant solution (e.g. lacking converters, unsuccessful power sources, etc....) and selects the suitable configuration according to the lowest NPC [22].

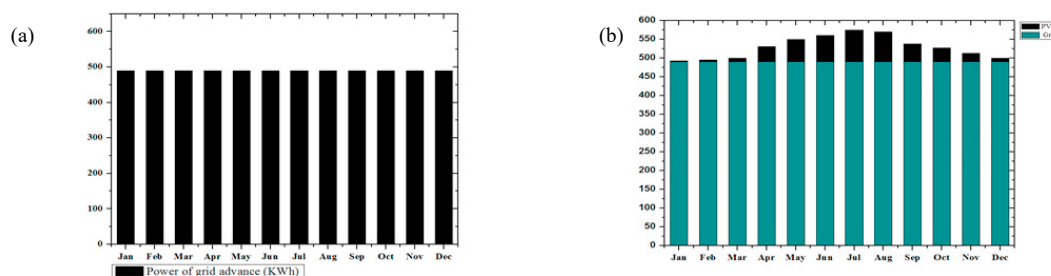


Fig. 5. (a) power production of grid (kWh); (b) comparison of power production between grid and PV panels.

It can be noticed from this result that the first configuration consists of solar PV panels, converter and advance grid which is the suitable configuration after optimization, but the energy supply directly transferred to the grid. In this paper, the second configuration result has been considered as optimum configuration with storage due to its reliability and that would be economical configuration. The energy amount is higher than other cases. The excess energy is positive which means that the microgrid has to supply the battery storage In Fig. 4. (a) On the cloudy

January and February, the solar PV generates lower levels of energy whereas solar radiation on June, July and August has an important potential.

Table 3. Optimization result –with grid.

case	Microgrid plan	Architecture				Cost		System	
		PV (KW)	Battery	Advance grid	Converter	NPC (M\$)	COE (\$)	Electric production (KWh/yr.)	Electric consumption (KWh/yr.)
Case 1		12	--	999.999	32	9.39	0.187	3929.86	3884.75
Case 2		12	17	999.999	32	9.41	0.187	3929.86	3884.75
Case 3		--	--	999.999	--	9.53	0.190	3878.6	3878.60
Case 4		--	20	999.999	8	9.54	0.190	3878.6	3878.6

The energy produced supplies the peak load demand. Fig. 4 (b) illustrates a typical monthly evolution of PV solar power output. It can be observed from the plot that the curve decays for January and September caused by the diversity of the atmospheric component like clouds, humidity etc. Otherwise, the increase energy output for March, April, May, June and July are performed, the amount energy of PV power output is close to 98 KW, where the lowest value is 15 KW. The monthly mean electric power production of PV system remains almost the same with slight variation of maximum and minimum of 557 KW and 456 KW in July and January, respectively, as shown in Fig. 5.(a) The total power production generated by grid was found to be maximum during the year expect December. In Fig. 5(b) a mean power production comparison between grid and PV panels. As we can see, the excess energy is higher especially in May, Jun, July and August due the solar potential.

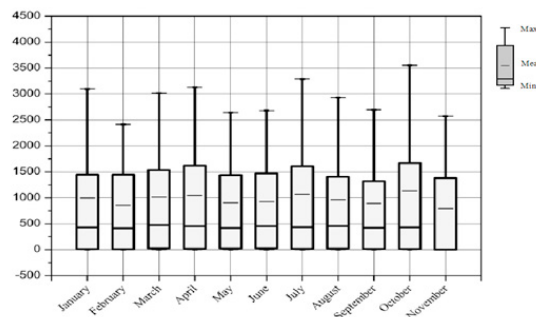


Fig. 6. Monthly average electric load (KW).

For the same optimum case, the energy load served for each month in a year is illustrated in Fig.6 It is obvious that in the months where PV energy is high, the state of charge of the battery maintains high value. Its value reaches the maximum allowable depth of discharge (the 30% level).

5.2. Sensitivity analysis

The main scenario considered in this study is a combination of PV panels, battery storage and a grid advance. For this scenario, the result illustrates that the lowest COE is 0.187 \$/KWh, the operating cost is 725.81 \$/year where the NPC of the optimal scenario is 9.41M\$ as shown in Table. 4. Any design process will have convenient results on the quality of input parameters in order to assess their effects on the overall specification of the global system design. The effect on COE due to variation in the life-time project, PV capital cost, the battery bank. It is clear from the

Table that the sensitivity analysis has been performed to assess the effect of solar radiation on the optimal system design and annualized energy consumption. The annual average solar radiation is close to 5.90 KWh/m²/day and the optimal size of PV arrays increase according to the cost and the operating of useful electric energy produced by the system.

Table 4. Sensitivity analysis cost.

Annual Average Solar Radiation (KWh/m ² /day)	PV (KW)	NPC(M\$)	COE (\$/KWh)	Operating Cost (\$/year)
4.58	12	9.41	0.187	725.812
4.73	25	9.45	0.188	729.069
5.90	45	9.50	0.189	729.338
5.60	50	9.54	0.190	733.757

6. Alternatives and proposed actions

HOMER software displays a list of optimized search space options based on the systems that yielded the lowest NPC as shown in Table.5. The table illustrates different components with various values contributed in this study, the PV panels capacity of the optimal design is 12 KW, the converter capacity is 32 KW. It is also to be noted that the capacity of PV panels is lower than the allowable limit of 10 % and 25 % to meet the demand optimally from its available resources as far as possible even when the unmet energy is relaxed.

Table 5. Search space.

Converter capacity (KW)	Converter dedicated (quantity)	PV panels capacity	Grid purchase capacity (KW)	Li-ASM strings
0	1	0	999999	0
8		12		2168
16		25		4336
24		45		6504
32		50		8672

6.1. Alternative 1: grid only

No action is required for this alternative. The existing electricity is purchased directly from the grid in accordance with the tariff, which in this case is 0.16 \$ / KWh. The renewable fraction of this alternative is zero percent.

6.2. Alternative 2: addition PV panels_ grid combination without storage

This option is viable and has an important potential to add more PV panels to the system, the electricity produced from the PV panels is combining directly with grid then the load demand is supplied without storage. This alternative appears to be reasonable from an economic point of view, where the price of electricity decreases in a regular way. The total investment is reduced as result to not deploying the batteries.

6.3. Alternative 3: addition PV panels_ grid combination with storage

The addition of storage batteries on the system will require a high capital investment. This possibility will be a slight increase in electricity price facing a deficit of energy .The combined systems are steadily into the network and improve power quality.

7. Conclusion

This paper deals with four scenarios that were analyzed and simulated with the platform HOMER software. The considered microgrid will be available at Abdelmalek Essaâdi University and the solar radiation is provided by energetic laboratory station on the same faculty. The configuration consists of PV panels, a battery storage, converter and advance grid which are considered to be economically feasible one. For the scenario, the COE is 0.187 \$/KWh, the NEP is 9.41M\$, which represents the purchasing cost of PV system. The operating cost is 725.812 \$/year representing the grid operation. The COE of each scenario is greater than purchasing energy from the grid. The load demand is largely fed by renewable sources with 10.63 MWh/year, which represent 74.3 % of the total energy production in microgrid. The implementation of microgrid system seems more accurate. The total investment of PV system and adoption of the feed in tariff will also encourage Abdelmalek Essaâdi University of Tetuan to install PV system in its building in the near future.

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